

T3 matched EFT Lagrangian terms: EFT_1_after_F

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out F. Remaining active T3 fields: S1, S2.

Field combinations 1 of 6

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	D	F	H
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$\begin{aligned} & +\frac{2}{3}\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{\omega^2}{g_{2,\tau_1,\tau_1}g_{2,\tau_2,\tau_2}S_{2,j_2}g_{2,\tau_1,\tau_2}g_{2,\tau_2,\tau_1}}\left(S_{2,j_1}^\dagger\right)^2 \\ & -\frac{2C\text{N e w S c a l a r }22\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{g_{2,\tau_1,\tau_1}^2S_{2,j_1}g_{2,\tau_1,\tau_1}S_{2,j_1}^\dagger}{g_{2,\tau_1,\tau_1}g_{2,\tau_2,\tau_2}S_{2,j_2}g_{2,\tau_1,\tau_2}g_{2,\tau_2,\tau_1}} \\ & -\frac{4}{3}\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{g_{2,\tau_1,\tau_1}^2S_{2,j_1}g_{2,\tau_1,\tau_1}S_{2,j_1}^\dagger}{g_{2,\tau_1,\tau_1}g_{2,\tau_2,\tau_2}S_{2,j_2}g_{2,\tau_1,\tau_2}g_{2,\tau_2,\tau_1}} \\ & -\frac{2\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{g_{2,\tau_1,\tau_1}^2S_{2,j_1}g_{2,\tau_1,\tau_1}S_{2,j_1}^\dagger}{g_{2,\tau_1,\tau_1}g_{2,\tau_2,\tau_2}S_{2,j_2}g_{2,\tau_1,\tau_2}g_{2,\tau_2,\tau_1}} \\ & -\frac{2\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{g_{2,\tau_1,\tau_1}^2S_{2,j_1}g_{2,\tau_1,\tau_1}S_{2,j_1}^\dagger}{g_{2,\tau_1,\tau_1}g_{2,\tau_2,\tau_2}S_{2,j_2}g_{2,\tau_1,\tau_2}g_{2,\tau_2,\tau_1}} \end{aligned}$	
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$\begin{aligned} & +\frac{2}{3}\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{\omega^2}{g_{1,\tau_1,\tau_1}g_{1,\tau_2,\tau_2}S_{1,j_2}g_{1,\tau_1,\tau_2}g_{1,\tau_2,\tau_1}}\left(S_{1,j_1}^\dagger\right)^2 \\ & -\frac{2C\text{N e w S c a l a r }12\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{g_{1,\tau_1,\tau_1}^2S_{1,j_1}g_{1,\tau_1,\tau_1}S_{1,j_1}^\dagger}{g_{1,\tau_1,\tau_1}g_{1,\tau_2,\tau_2}S_{1,j_2}g_{1,\tau_1,\tau_2}g_{1,\tau_2,\tau_1}} \\ & -\frac{4}{3}\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{g_{1,\tau_1,\tau_1}^2S_{1,j_1}g_{1,\tau_1,\tau_1}S_{1,j_1}^\dagger}{g_{1,\tau_1,\tau_1}g_{1,\tau_2,\tau_2}S_{1,j_2}g_{1,\tau_1,\tau_2}g_{1,\tau_2,\tau_1}} \\ & -\frac{2\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{g_{1,\tau_1,\tau_1}^2S_{1,j_1}g_{1,\tau_1,\tau_1}S_{1,j_1}^\dagger}{g_{1,\tau_1,\tau_1}g_{1,\tau_2,\tau_2}S_{1,j_2}g_{1,\tau_1,\tau_2}g_{1,\tau_2,\tau_1}} \\ & -\frac{2\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{g_{1,\tau_1,\tau_1}^2S_{1,j_1}g_{1,\tau_1,\tau_1}S_{1,j_1}^\dagger}{g_{1,\tau_1,\tau_1}g_{1,\tau_2,\tau_2}S_{1,j_2}g_{1,\tau_1,\tau_2}g_{1,\tau_2,\tau_1}} \end{aligned}$	

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	D	F	H
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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	$W_{\mu\nu}$	constant
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Field combinations 3 of 6

[illegible]

[illegible]

Field combinations 4 of 6

[illegible]

Field combinations 5 of 6

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	F	H
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0			
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0			
T3-A	0	1	0	3	1	2	$\frac{1}{2}$			
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{h \log \left(\frac{\mu b m_2^2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,\tau_1} \overline{x_{\tau_3,a_1,a_2,j_1}} \lambda_{\tau_3} S_2^\dagger}{\sqrt{3}}$		
								$-h \log \left(\frac{\mu b m_2^2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,\tau_1} \overline{x_{\tau_3,a_1,a_2,j_1}} \lambda_{\tau_3} S_2^\dagger$		
								$-2h \log \left(\frac{\mu b m_2^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} \overline{y_{e,i_2,i_2}} H_{a_1} S_2 y_{2,i_2,\tau_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger$		
								$-\frac{h \log \left(\frac{\mu b m_2^2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} \overline{\lambda_{\tau_3}} S_2 y_{1,i_1,\tau_1} \text{Bn r(T3 Mix CG)}_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$		
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-h \log \left(\frac{\mu b m_2^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} \overline{\lambda_{\tau_3}} S_2 y_{2,i_1,\tau_1} \text{Bn r(T3 Mix CG)}_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$		
								$-2h \log \left(\frac{\mu b m_2^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} H_{a_2} S_2 y_{2,i_1,\tau_1} H_{a_1}^\dagger S_2^\dagger \overline{x_{\text{H2},a_1,a_2}^1} \lambda_{\text{H2}^1}$		